

APPLIED DATA SCIENCE CAPSTONE

SpaceX Falcon 9 First-Stage Landing Prediction

Predicting first-stage recovery success from launch telemetry, using data collected via public API and web scraping, then modeled with classical ML classifiers.

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GitHub Repository



All notebooks, Python scripts, and SQL files for this project are available at:

<https://github.com/jai1116-lang/spacex-falcon9-capstone>

- Data collection notebooks (API + web scraping)
- Data wrangling & labeling scripts
- EDA (SQL + visualization) notebooks
- Interactive visual analytics (Folium, Plotly Dash)
- Predictive analysis / classification notebook

Executive Summary

SpaceX advertises Falcon 9 launches at ~\$62M, versus \$165M+ for competitors — largely because SpaceX reuses the first stage. If we can predict whether a landing will succeed before launch, we can estimate launch cost and better evaluate a competing bid.

90

Launches analyzed

Falcon 9 only, Falcon 1 excluded

66.7%

Historical success rate

First-stage recovery, all sites

83.3%

Best model test accuracy

Decision Tree classifier

3

Launch sites

CCAFS, KSC, VAFB

Key finding

Landing success correlates strongly with orbit type and launch recency: later flights (higher flight number), lighter payloads, and orbits like VLEO/LEO land more reliably than heavy GTO missions.

Project Background & Questions

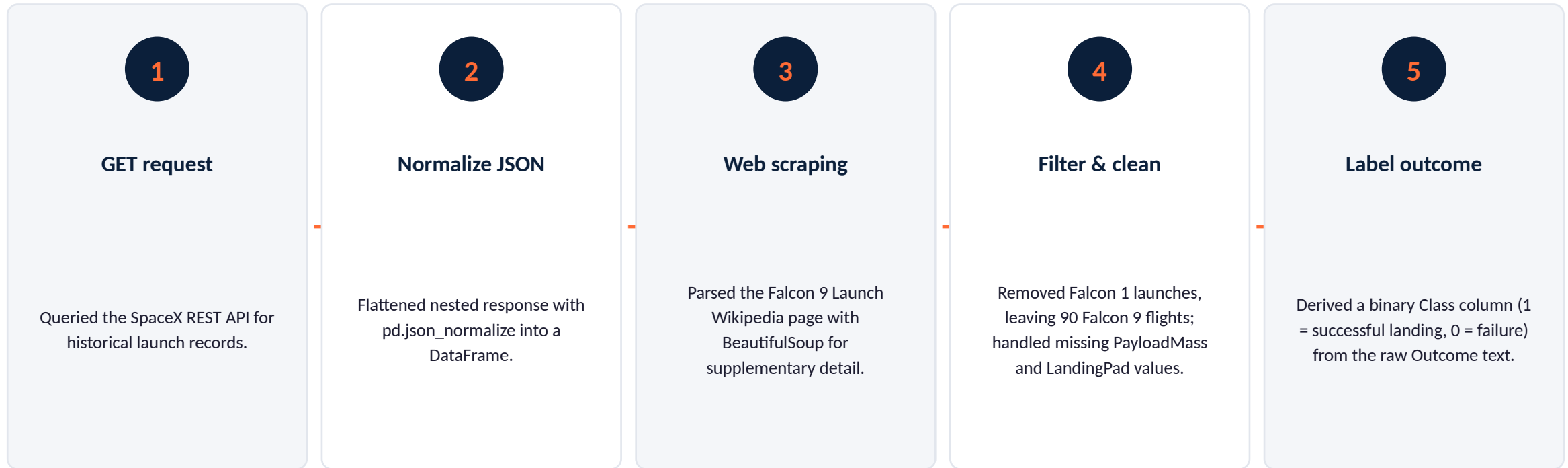
Background

SpaceX's reusable Falcon 9 first stage is the key driver of its low launch cost. Whether a first stage lands successfully depends on many factors — payload mass, orbit, launch site, booster generation, and more. This project builds a data pipeline and classification model to predict landing outcome from pre-launch data.

Questions this project answers

- 1 What features most influence landing success?
- 2 Does landing success improve over time / mission number?
- 3 Which launch sites and orbits have the best track record?
- 4 Can we predict landing outcome with a classifier — and how accurate is it?

Data Collection & Wrangling



Result: a clean, labeled dataset of 90 Falcon 9 launches with 17 columns spanning payload, orbit, site, booster, and landing outcome.

EDA & Interactive Visual Analytics



Visual EDA

Seaborn/Matplotlib scatter & bar plots:
FlightNumber vs. PayloadMass,
LaunchSite success rate, Orbit vs.
Outcome, yearly trend.



SQL EDA

Loaded launch data into Db2 and
queried launch-site counts, payload
aggregates, and outcome breakdowns
with SQL.



Folium map

Plotted launch sites and individual
launch outcomes (success/fail markers)
on an interactive map, with distances to
coastline/city.



Plotly Dash

Built an interactive dashboard with a
launch-site dropdown and payload-
range slider driving a pie chart and
scatter plot.

Predictive Analysis (Classification)

1

Standardize features (StandardScaler) after one-hot encoding categoricals (Orbit, LaunchSite, LandingPad, Serial, GridFins, Reused, Legs) — 83 total columns.

2

Split into train/test sets (80/20), yielding 18 test samples.

3

Grid-search four classifiers — Logistic Regression, SVM, Decision Tree, KNN — with 10-fold cross-validation on the training set.

4

Select best hyperparameters per model by validation accuracy, then evaluate all four on the held-out test set.

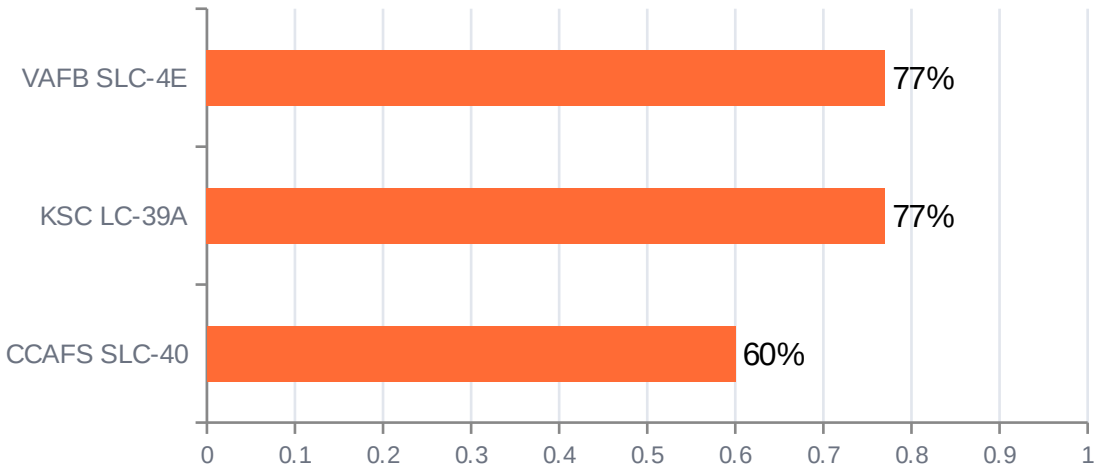
5

Compare test accuracy and confusion matrices to choose the final model.

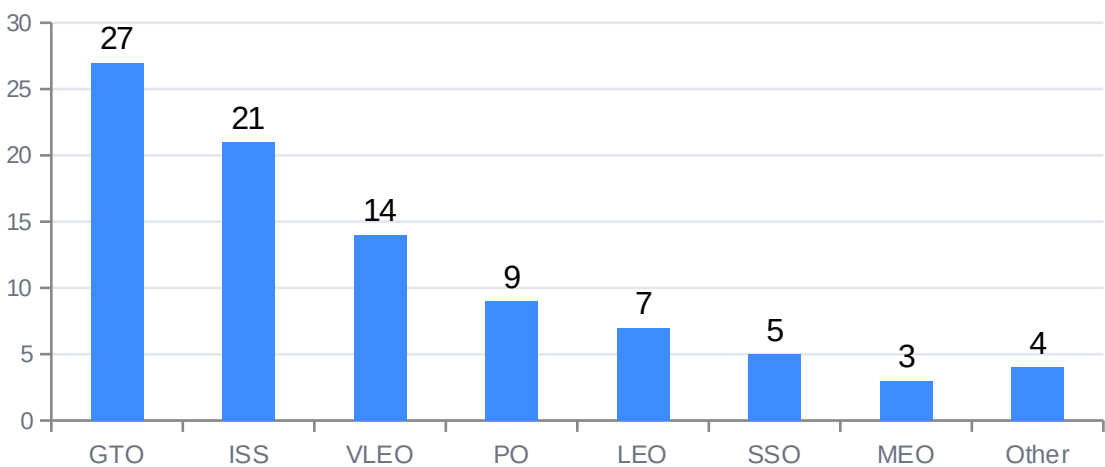
RESULTS

EDA — Visualization

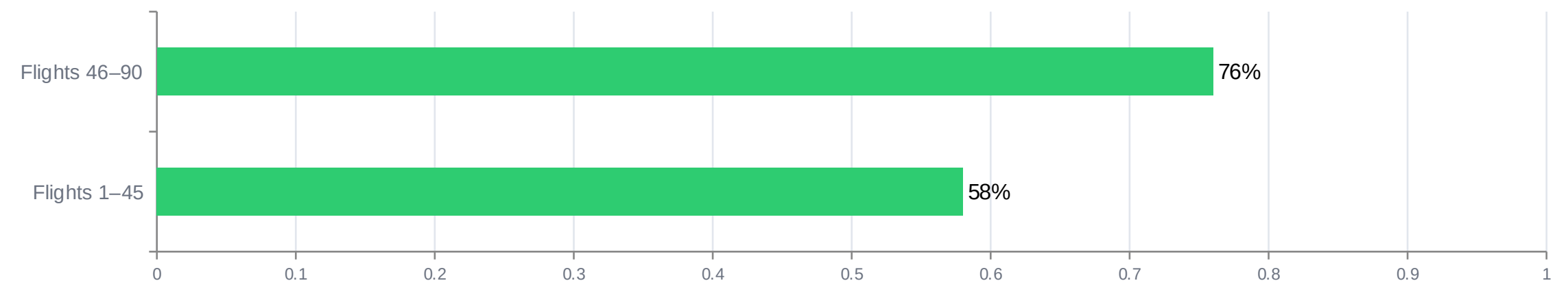
Success rate by launch site



Launch count by orbit type



Landing success improves with mission experience



RESULTS

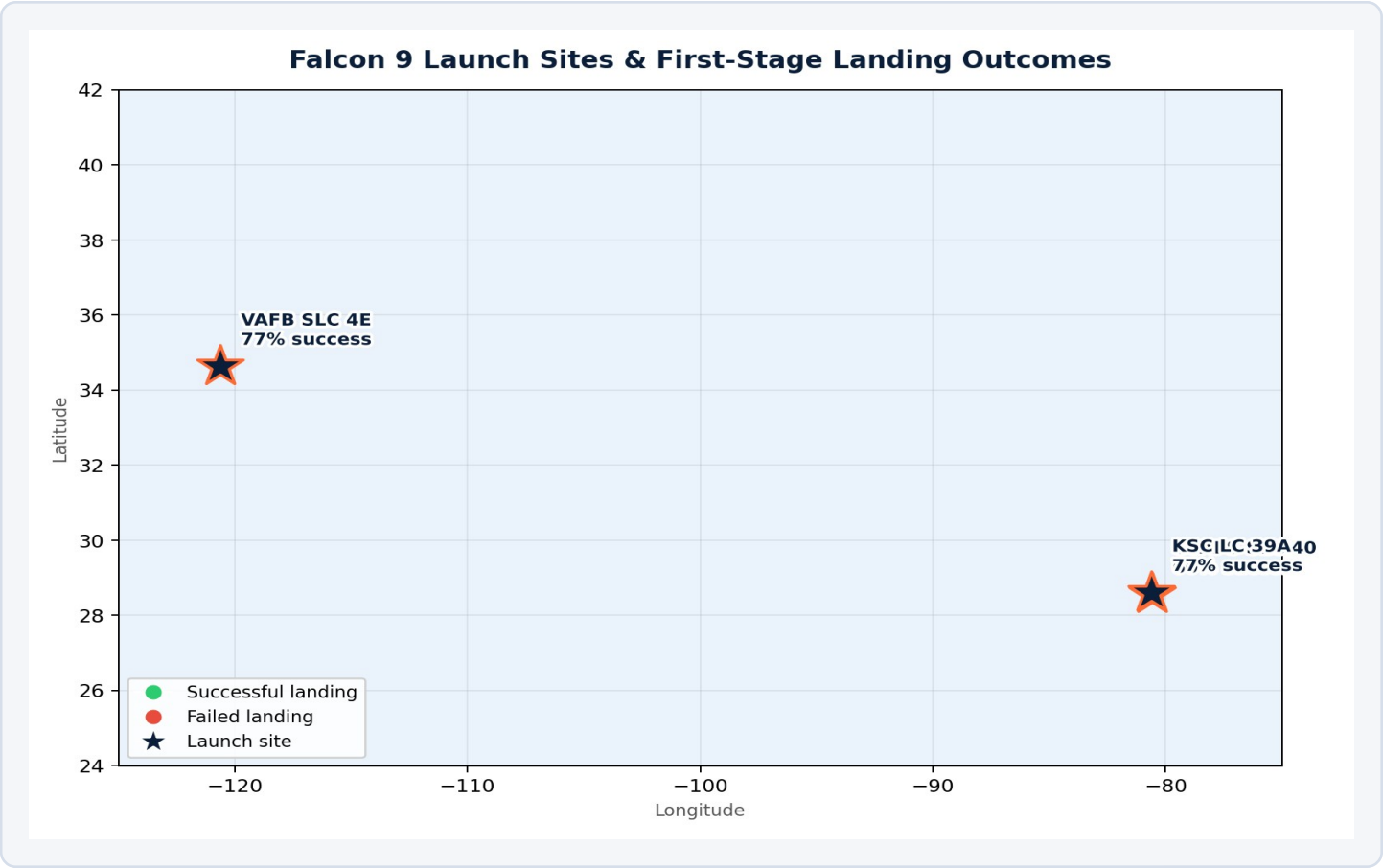
EDA — SQL

SQL Query	Result
SELECT COUNT(*) FROM SPACEXTBL WHERE LaunchSite = 'CCAFS SLC 40'	55 launches
SELECT SUM(Class)*1.0/COUNT(*) FROM SPACEXTBL	66.7% success (60 of 90)
SELECT COUNT(*) FROM SPACEXTBL WHERE Orbit = 'GTO'	27 launches
SELECT COUNT(*) FROM SPACEXTBL WHERE Outcome = 'True ASDS'	41 drone-ship landings
SELECT DISTINCT LaunchSite FROM SPACEXTBL	CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
SELECT MIN(PayloadMass) FROM SPACEXTBL	350 kg

Full query set (9 queries) is in `sql/eda_queries.sql` and `notebooks/03_eda_sql.ipynb` in the GitHub repo.

RESULTS

Interactive Map — Folium

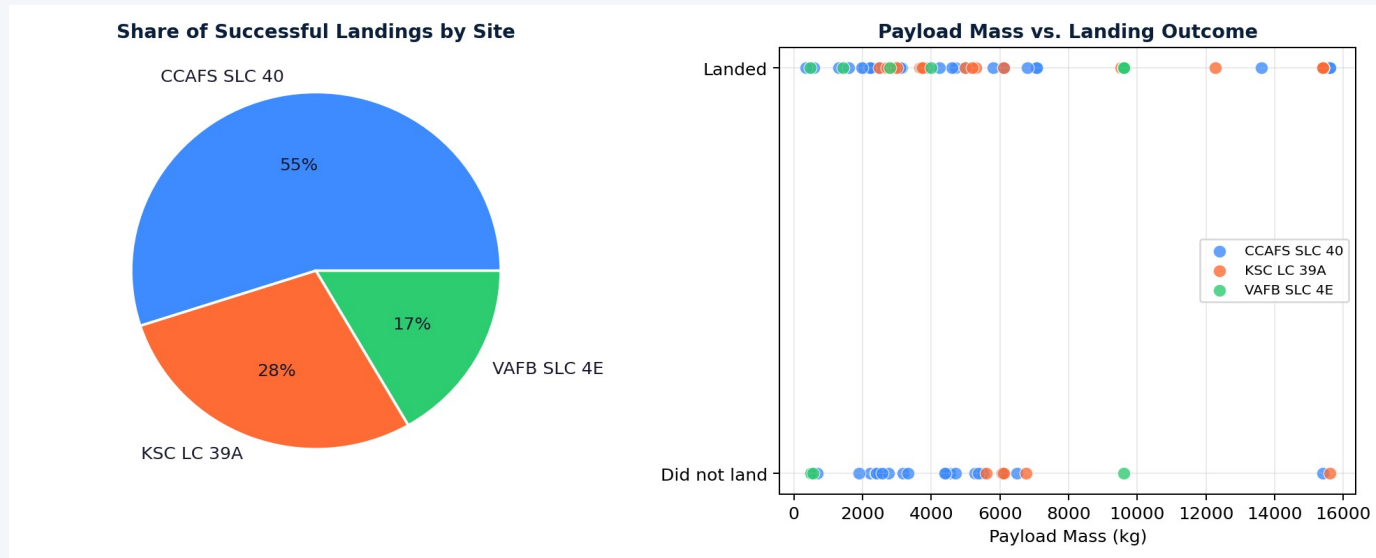


What the map shows

- Markers for all 3 launch sites, colored by aggregate success rate.
- Individual launch markers (green = success, red = failure) clustered per site.
- Distance lines from each site to the nearest coastline, railway, and city — used to assess siting constraints.
- Confirms launch sites are built near the coast, away from population centers, consistent with safety practice.

RESULTS

Interactive Dashboard — Plotly Dash



Static preview — the live app (notebook 5) adds an interactive site dropdown and payload-mass range slider that filter both charts.

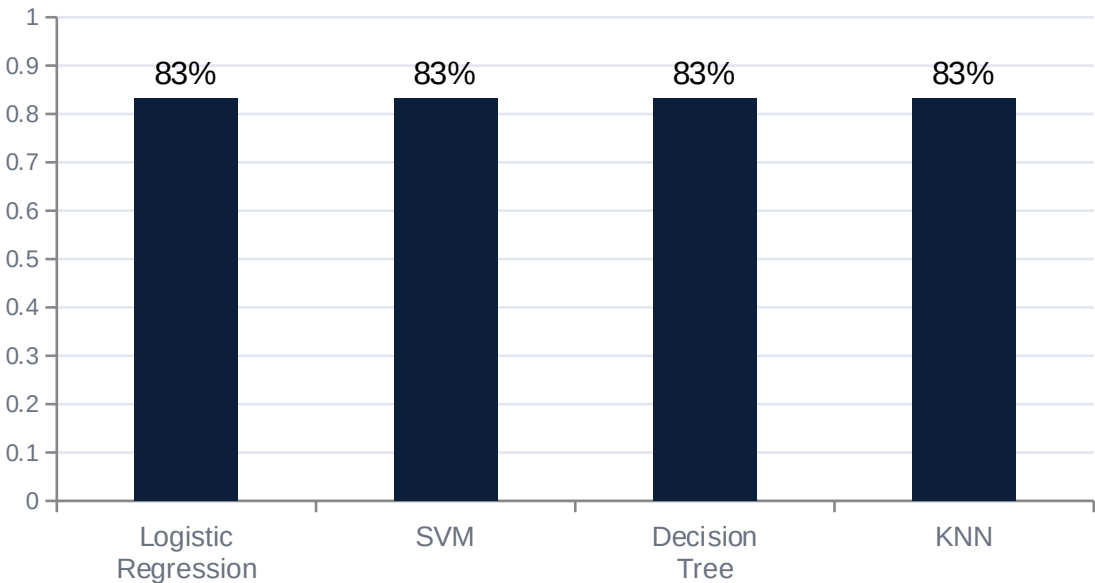
Dashboard features

- Dropdown to filter by launch site or view all sites combined.
- Pie chart of successful landings by site — KSC LC-39A leads at ~77%.
- Range slider to filter by payload mass (kg), updating the scatter plot live.
- Scatter plot of payload mass vs. landing outcome, colored by booster version.

RESULTS

Predictive Analysis — Classification

Test-set accuracy by model



All four models converge near the same test accuracy on this small (18-sample) held-out set — differences of 1–2 predictions shift the ranking, so cross-validation accuracy (below) is the more stable comparison.

Best model — Decision Tree

Metric	Value
Test accuracy	83.3%
Best criterion	gini
Best max_depth	6
Test samples	18 (20% holdout)

Confusion matrix (Decision Tree, test set)

	Pred: Fail	Pred: Land
Actual: Fail	5	1
Actual: Land	2	10

CONCLUSION

Key Takeaways

1

Landing success is learnable. A Decision Tree classifier reached 83.3% test accuracy predicting first-stage landing outcome from pre-launch features.

2

Experience matters. Success rate rose from ~58% in the first 45 launches to ~76% in the most recent 45 — SpaceX's landing process improved measurably over time.

3

Orbit and site matter. KSC LC-39A and VAFB SLC-4E outperform CCAFS SLC-40 (~77% vs. 60%); heavy GTO missions land less reliably than lighter LEO/VLEO missions.

4

Practical value. This kind of model lets a buyer estimate whether a quoted Falcon 9 launch will reuse its booster — and therefore estimate the likely price — before the mission flies.

Next steps: incorporate weather data, add booster-generation features, and test ensemble methods (Random Forest, gradient boosting) for a further accuracy gain.